

Real Estate DevOps Project Part-1

Goal: Build a production-style Flask application using Linux, Git, Docker, AWS and Terraform, then extend it with Jenkins, Ansible and CloudWatch.

1. Project Structure

```
cd /real-estate-devops
tree -L 2
```

```
root@psr-app:~# cd real-estate-devops/
root@psr-app:~/real-estate-devops# tree -L 2
```

```
.
├── ansible
├── app
│   ├── Dockerfile
│   ├── app.py
│   ├── requirements.txt
│   └── venv
├── docs
├── jenkins
├── scripts
└── terraform
    ├── alb.tf
    ├── aws
    ├── awsclicv2.zip
    ├── ec2.tf
    ├── internet-gateway.tf
    ├── outputs.tf
    ├── provider.tf
    ├── rds.tf
    ├── route-tables.tf
    ├── security-groups.tf
    ├── subnets.tf
    ├── terraform.tfstate
    ├── terraform.tfvars
    ├── variables.tf
    ├── versions.tf
    └── vpc.tf
```

```
9 directories, 18 files
```

```
root@psr-app:~/real-estate-devops#
```

2. Flask Application

```
cd ~/real-estate-devops/app
source venv/bin/activate
```

```
python3 app.py
```

```
(venv) root@psr-app:~/real-estate-devops/app# source venv/bin/activate
```

```
python3 app.py
```

```
* Serving Flask app 'app'
```

```
* Debug mode: off
```

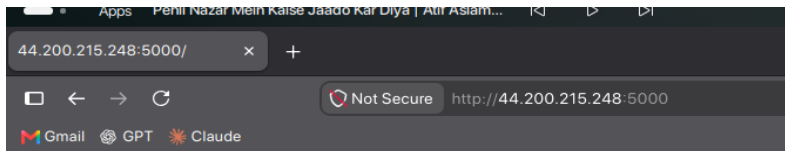
```
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
```

```
* Running on all addresses (0.0.0.0)
```

```
* Running on http://127.0.0.1:5000
```

```
* Running on http://172.31.3.57:5000
```

```
Press CTRL+C to quit
```



Real Estate Application

Version: 1.0

Environment: Development

3. Git / GitHub

```
cd ~/real-estate-devops
git status
git log --oneline -3
git remote -v
```

```
root@psr-app:~# cd real-estate-devops/
root@psr-app:~/real-estate-devops# git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   ../snap/amazon-ssm-agent/current

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    ../.aws/
    ../.bash_history
    ../.gitconfig
    ../.lessshst
    ../.ssh/known_hosts
    ../.ssh/known_hosts.old
    ../.terraform.d/
    ../.viminfo
    ../.wget-hsts
    ../one-key.pem
    .dockerignore
    .gitignore
    .terraform.lock.hcl
    app/Dockerfile
    terraform/

no changes added to commit (use "git add" and/or "git commit -a")
root@psr-app:~/real-estate-devops#
```

```
root@psr-app:~/real-estate-devops# git log --oneline -3
9a58fb3 (HEAD -> main, origin/main) Initial application
root@psr-app:~/real-estate-devops#
```

```
root@psr-app:~/real-estate-devops# git remote -v
origin https://github.com/jpz6/real-estate-devops.git (fetch)
origin https://github.com/jpz6/real-estate-devops.git (push)
root@psr-app:~/real-estate-devops#
```

4. Docker

```
cd ~/real-estate-devops
docker images
docker ps
```

```
root@psr-app:~# cd real-estate-devops/
root@psr-app:~/real-estate-devops# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
python:3.12-slim	f77ac9e44ae9	179MB	45.4MB	
real-estate-app:1.0	50a225b1ea78	198MB	48.2MB	U

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
76ca3172da8e	real-estate-app:1.0	"python app.py"	36 hours ago	Up 36 hours	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp

```
root@psr-app:~/real-estate-devops#
```

5. Terraform

```
cd ~/real-estate-devops/terraform
terraform fmt -check
terraform validate
terraform plan

root@psr-app:~/real-estate-devops/terraform# terraform validate
Success! The configuration is valid.

root@psr-app:~/real-estate-devops/terraform# terraform plan
random_password.db: Refreshing state... [id=none]
data.aws_availability_zones.available: Read complete after 0s [id=us-east-1]
data.aws_ami.ubuntu: Reading...
aws_vpc.main: Refreshing state... [id=vpc-0014cd7d4e335e5d6]
data.aws_availability_zones.available: Read complete after 0s [id=us-east-1]
data.aws_ami.ubuntu: Read complete after 0s [id=ami-0045d7fc2ad003464]
aws_subnet.private_b: Refreshing state... [id=subnet-0ba876f9195da4e5e]
aws_route_table.private: Refreshing state... [id=rtb-0299522700abba19d]
aws_security_group.alb: Refreshing state... [id=sg-0eca30c7bdd58d139]
aws_subnet.public_b: Refreshing state... [id=subnet-0b5264ee1b99eb7c0]
aws_subnet.public_a: Refreshing state... [id=subnet-0601b0c3f7d6f24fc]
aws_internet_gateway.main: Refreshing state... [id=igw-06c1ae848404f0a9e]
aws_subnet.private_a: Refreshing state... [id=subnet-05f72c351cfe01164]
aws_lb_target_group.app: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:860668857234:targetgroup/real-estate-tg/7347f0b5830157fd]
aws_route_table.public: Refreshing state... [id=rtb-0b549441bceceb571]
aws_security_group.app: Refreshing state... [id=sg-035c2805048b03493]
aws_route_table_association.private_b: Refreshing state... [id=rtbassoc-081302c8b40428f3]
aws_route_table_association.private_a: Refreshing state... [id=rtbassoc-0ac86f9d9510057e]
aws_db_subnet_group.rds: Refreshing state... [id=real-estate-db-subnet-group]
aws_lb.app: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:860668857234:loadbalancer/app/real-estate-alb/8da25022a5766c4e]
aws_route_table_association.public_a: Refreshing state... [id=rtbassoc-0781b413652126d4c]
aws_route_table_association.public_b: Refreshing state... [id=rtbassoc-022e42afd26f62852]
aws_security_group.app: Refreshing state... [id=sg-04fbb1aecd45c5dfb6]
aws_instance.app: Refreshing state... [id=i-05e83bce58e2b1f87]
aws_lb_listener.http: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:860668857234:listener/app/real-estate-alb/8da25022a5766c4e/4b92f957926b3ea8]
aws_db_instance.postgres: Refreshing state... [id=db-TEFNG65EUNPK3MFWCUZ6VNN57Y]
aws_lb_target_group_attachment.app: Refreshing state... [id=arn:aws:elasticloadbalancing:us-east-1:860668857234:targetgroup/real-estate-tg/7347f0b5830157fd,1-05e83bce58e2b1f87,5000]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
root@psr-app:~/real-estate-devops/terraform#
```

terraform output

```
root@psr-app:~/real-estate-devops/terraform#
root@psr-app:~/real-estate-devops/terraform#
root@psr-app:~/real-estate-devops/terraform# terraform output
alb_dns_name = "real-estate-alb-537300692.us-east-1.elb.amazonaws.com"
app_instance_id = "i-05e83bce58e2b1f87"
app_private_ip = "10.0.1.241"
app_public_ip = "52.71.119.186"
db_password = <sensitive>
rds_endpoint = "real-estate-postgres.cy94ca2uwa3o.us-east-1.rds.amazonaws.com"
rds_port = 5432
vpc_id = "vpc-0014cd7d4e335e5d6"
root@psr-app:~/real-estate-devops/terraform#
```

6. AWS Infrastructure

Show: VPC → Subnets → Security Groups → EC2 → ALB → RDS.

AWS resources

Choose filter set

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	real-estate-ap...	i-05e83bce58e2b1f87	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-52-71-119-186.co...	52.71.119.186	-	-
☐	psr-app	i-02d7ed4ad00232912	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-44-200-215-248.co...	44.200.215.248	-	-

ALB

Q Filter load balancers

☒	Name ▾	State ▾	Type ▾	Scheme ▾	IP address type ▾	VPC ID ⓘ ▾	Availability Zones ▾	Security groups ▾	DNS name
☒	real-estate-alb	Active	application	Internet-facing	IPv4	vpc-0014cd7d4e335e5d6	2 Availability Zones	sg-0eca30c7bdd58d13...	real-esta

Target Groups

Target groups (1) [Info](#) | [What's new?](#)

☐	Name ▾	ARN ▾	Port ▾	Protocol ▾	Target type ▾	Load balancer ▾	VPC ID
☐	real-estate-tg	 arn:aws:elasticloadbalancin...	5000	HTTP	Instance	real-estate-alb	vpc-0014cd7d4e335e5d6

Security Groups

real-estate-alb-sg	sg-0eca30c7bdd58d139	real-estate-alb-sg	vpc-0014cd7d4e335e5d6 ⓘ	Security group for Application Load Bal...	860668857234	1 Permission entry
real-estate-app-sg	sg-035c2805048b03493	real-estate-app-sg	vpc-0014cd7d4e335e5d6 ⓘ	Security group for application EC2	860668857234	2 Permission entries
real-estate-rds-sg	sg-04fbb1aecd45c5dfb6	real-estate-rds-sg	vpc-0014cd7d4e335e5d6 ⓘ	Security group for RDS	860668857234	1 Permission entry

VPC

☐ Name ▾	VPC ID ▾	State ▾	Encryption c... ▾	Encryption control ... ▾	Block Public... ▾	IPv4 CIDR ▾	IPv6 CIDR ▾	DHCP option set ▾
☐ real-estate-vpc	vpc-0014cd7d4e335e5d6	Available	–	–	Off	10.0.0.0/16	–	dopt-030f95a0fb5ff48bb
☐ default-vpc-do-not-delete	vpc-0e1ad5848a7c7982b	Available	–	–	Off	172.31.0.0/16	–	dopt-030f95a0fb5ff48bb

Subnets

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association ID	Avail
☐	real-estate-public-b	subnet-0b5264ee1b99eb7c0	Available	vpc-0014cd7d4e335e5d6 real-estate-private-a	Off	10.0.2.0/24	-	-	250
☐	real-estate-private-a	subnet-05f72c351cfe01164	Available	vpc-0014cd7d4e335e5d6 real-estate-private-b	Off	10.0.10.0/24	-	-	250
☐	real-estate-public-a	subnet-0691b0c3f7d6f24fc	Available	vpc-0014cd7d4e335e5d6 real-estate-private-a	Off	10.0.1.0/24	-	-	249
☐	real-estate-private-b	subnet-0ba876f9195da4e5e	Available	vpc-0014cd7d4e335e5d6 real-estate-private-b	Off	10.0.20.0/24	-	-	251

Route Tables

☐	real-estate-private-rt	rtb-02995227d0abba19d	2 subnets	-	No	vpc-0014cd7d4e335e5d6 real-estate-private-a	860668857234
☐	real-estate-public-rt	rtb-0b549441bccecb571	2 subnets	-	No	vpc-0014cd7d4e335e5d6 real-estate-private-a	860668857234

IGW

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	lgw-02aee6a11976a9bb7	Attached	vpc-0e1ad5848a7c7982b default-vpc-	860668857234
☐	real-estate-igw	lgw-06c1ae848404f0a9e	Attached	vpc-0014cd7d4e335e5d6 real-estate-private-a	860668857234

NACL

☐	Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbo
☐	-	acl-0c26cc7f9bdeb29c2	4 Subnets	Yes	vpc-0014cd7d4e335e5d6 / real-estate-private-a	2 Inbound rules	2 Outb

DB

☐	DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendations	CPU	Current activity
☐	real-estate-postgres	Available	Instance	PostgreSQL	SECOND	us-east-1a	db.t3.micro		4.90%	0 Connections